



The Bitcoin Whitepaper in 2 Pages

The plain-English gist of Satoshi Nakamoto's 2008 paper, "Bitcoin: A Peer-to-Peer Electronic Cash System."

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"A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution."

Abstract, first sentence
bitcoin.org/bitcoin.pdf ↗

1 The problem: double spending

Digital money is just data, and data can be copied. What stops someone from sending the same coin to two people? Until 2008 the answer was a trusted middleman, like a bank, keeping the official ledger. The paper asks for "an electronic payment system based on cryptographic proof instead of trust."

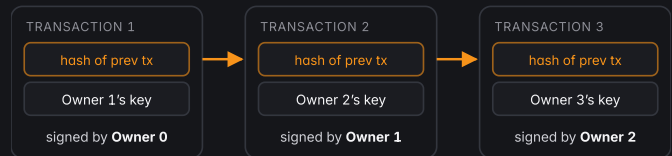


Satoshi's key point: the only way to know a coin wasn't already spent is for everyone to see every transaction, in order.

2 Coins are a chain of signatures

The paper puts it simply: "We define an electronic coin as a chain of digital signatures." To pay you, the current owner signs a message that points to the previous transaction and names your public key. Anyone can check the signatures. Nobody can forge them without the owner's private key.

Each payment points back to the one before and names the next owner.



3 A shared clock: blocks, hashes and proof-of-work

Each block carries the hash of the one before it. Change one and every later link breaks.



Transactions are bundled into **blocks**. Each block includes the **hash** (a digital fingerprint) of the block before it, so the blocks form a chain in time order.

To add a block, a computer must find a **nonce** that gives the block a hash below a target. That takes real computing work and electricity to find, but is instant to check. This is **proof-of-work**.

To rewrite history, an attacker would have to redo that work for the changed block and every block after it, and outrun everyone else.

How the network runs, in the paper's six steps

- 1** New transactions are broadcast to all nodes
- 2** Each node collects them into a block
- 3** Each node works on the proof-of-work
- 4** The winner broadcasts its block
- 5** Nodes accept it only if every transaction is valid and unspent
- 6** Nodes build the next block on top of it

The whole paper in one paragraph

Bitcoin replaces the bank's ledger with a public one that everyone keeps a copy of. Coins move by digital signature. Transactions are grouped into blocks, and each block is locked to the one before it by a hash and by real computing work. The honest majority of computing power always extends the longest chain, so rewriting the past gets harder with every block. Miners are paid in new coins and fees for doing this work, which makes playing by the rules the profitable choice.

Key words

Hash a fixed-length fingerprint of data · **Node** a computer running the network rules · **Miner** a node doing proof-of-work · **Block** a batch of transactions · **Public / private key** the key others see / the secret that signs

